

**Table 1. compares the prime-counting error at powers of ten, the difference between  $\text{Li}(10^n)$  and  $\pi(10^n)$ , and the candidate term  $C_{\text{comp}}$  proposed in this study, for  $n = 1-25$ .**

| $10^n$    | $\pi(10^n)$                     | $C_{\text{cand}}(n)$    | $\pi(10^n)-4$                   | $C_{\text{comp}}$                 | $\text{Li}-\pi$ |
|-----------|---------------------------------|-------------------------|---------------------------------|-----------------------------------|-----------------|
| $10^1$    | 4                               | 0                       | 0                               | 0                                 | 2               |
| $10^2$    | 25                              | $(8 \div 3)(10^1-1)$    | 21                              | 3                                 | 5               |
| $10^3$    | 168                             | $(8 \div 3)(10^2-1)$    | 164                             | 100                               | 10              |
| $10^4$    | 1,229                           | $(8 \div 3)(10^3-1)$    | 1,225                           | 1,439                             | 17              |
| $10^5$    | 9,592                           | $(8 \div 3)(10^4-1)$    | 9,588                           | 17,076                            | 38              |
| $10^6$    | 78,498                          | $(8 \div 3)(10^5-1)$    | 78,494                          | 188,170                           | 130             |
| $10^7$    | 664,579                         | $(8 \div 3)(10^6-1)$    | 664,575                         | 2,002,089                         | 339             |
| $10^8$    | 5,761,455                       | $(8 \div 3)(10^7-1)$    | 5,761,451                       | 20,905,213                        | 754             |
| $10^9$    | 50,847,534                      | $(8 \div 3)(10^8-1)$    | 50,847,530                      | 215,819,134                       | 1,701           |
| $10^{10}$ | 455,052,511                     | $(8 \div 3)(10^9-1)$    | 455,052,507                     | 2,211,614,157                     | 3,103           |
| $10^{11}$ | 4,118,054,813                   | $(8 \div 3)(10^{10}-1)$ | 4,118,054,809                   | 22,548,611,855                    | 11,588          |
| $10^{12}$ | 37,607,912,018                  | $(8 \div 3)(10^{11}-1)$ | 37,607,912,014                  | 229,058,754,650                   | 38,263          |
| $10^{13}$ | 346,065,536,839                 | $(8 \div 3)(10^{12}-1)$ | 346,065,536,835                 | 2,320,601,129,829                 | 108,971         |
| $10^{14}$ | 3,204,941,750,802               | $(8 \div 3)(10^{13}-1)$ | 3,204,941,750,798               | 23,461,724,915,866                | 314,890         |
| $10^{15}$ | 29,844,570,422,669              | $(8 \div 3)(10^{14}-1)$ | 29,844,570,422,665              | 236,822,096,243,999               | 1,052,619       |
| $10^{16}$ | 279,238,341,033,925             | $(8 \div 3)(10^{15}-1)$ | 279,238,341,033,921             | 2,387,428,325,632,743             | 3,214,632       |
| $10^{17}$ | 2,623,557,157,654,233           | $(8 \div 3)(10^{16}-1)$ | 2,623,557,157,654,229           | 24,043,109,509,012,435            | 7,956,589       |
| $10^{18}$ | 24,739,954,287,740,860          | $(8 \div 3)(10^{17}-1)$ | 24,739,954,287,740,856          | 241,926,712,378,925,808           | 7,927,346       |
| $10^{19}$ | 234,057,667,276,344,607         | $(8 \div 3)(10^{18}-1)$ | 234,057,667,276,344,603         | 2,432,608,999,390,322,061         | 99,877,775      |
| $10^{20}$ | 2,220,819,602,560,918,840       | $(8 \div 3)(10^{19}-1)$ | 2,220,819,602,560,918,836       | 24,445,847,064,105,747,828        | 4,583,891       |
| $10^{21}$ | 21,127,269,486,018,731,928      | $(8 \div 3)(10^{20}-1)$ | 21,127,269,486,018,731,924      | 245,539,397,180,647,934,740       | 1,157,446,844   |
| $10^{22}$ | 201,467,286,689,315,906,290     | $(8 \div 3)(10^{21}-1)$ | 201,467,286,689,315,906,286     | 2,465,199,379,977,350,760,378     | 193,519,524     |
| $10^{23}$ | 1,925,320,391,606,803,968,923   | $(8 \div 3)(10^{22}-1)$ | 1,925,320,391,606,803,968,919   | 24,741,346,275,059,862,697,745    | 1,215,343,115   |
| $10^{24}$ | 18,435,599,767,349,200,867,866  | $(8 \div 3)(10^{23}-1)$ | 18,435,599,767,349,200,867,862  | 248,231,066,899,317,465,798,802   | 1,714,694,528   |
| $10^{25}$ | 176,846,309,399,143,769,411,680 | $(8 \div 3)(10^{24}-1)$ | 176,846,309,399,143,769,411,676 | 2,489,820,357,267,522,897,254,988 | 5,516,098,882   |